

2.2 Counter-example

Consider a purely supersonic case for the 2D MHD equation: $\rho \equiv 1$, $\mathbf{m} \equiv (1, 1)$, $\mathbf{B} = A(x, -y)$ where $|A| \leq \delta$. Then these quantities can be well-resolved by the lowest-order GDF-DG scheme. Consider the main element $K = [0, h]^2$ (where we analyse the evolution). We assume that in all of its neighbour elements (including the corner ones) the pressure is small enough and the global constant δ is also small enough so that the information is purely propogating to the top-right direction, which implies the upwind flux is the exact flux. Moreover, now the true physical signal speed is bounded uniformly, say $\sigma \leq \sigma_{\max}$ ($\sigma_{\max} \geq 1$), then the CFL condition should be like $\frac{2\Delta t}{h} \leq \frac{\text{CFL}}{\sigma_{\max}}$. Use 2-point Gauss-Lobatto for all quadratures. Then, in order to consider the evolution on K , it suffices to assign initial (piecewise constant) total energy E for only the region $[-h, h]^2$. We assign it to be $1 + A^2 h^2 + \epsilon_j$ (j for the index of the quadrangle) Then the semi-discrete scheme gives (within K)

$$\rho(t) \equiv 1, \quad (2.6a)$$

$$\mathbf{B}(x, y, t) = A(x - t, t - y), \quad (2.6b)$$

$$\mathbf{m}(t) = (1 + \frac{t(\gamma - 1)}{2h}(A^2 h^2 - (\epsilon_1 - \epsilon_2 - \epsilon_3 + \epsilon_4)), 1 + \frac{t(\gamma - 1)}{2h}(A^2 h^2 - (\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4))), \quad (2.6c)$$

$$E(t) = 1 + A^2 h^2 + \epsilon_1 + \frac{t(\gamma - 1)}{2h} 2A^2 h^2 - \frac{t\gamma}{2h} (2\epsilon_1 - 2\epsilon_3). \quad (2.6d)$$

At time $\Delta t = 2h\lambda$ (assume $\lambda \leq \min\left\{\frac{1}{4}, \frac{\text{CFL}}{4\sigma_{\max}}\right\}$) so that the maximum of $|B|^2$ is achieved at $(x, t) = (h, h)$, the internal energy at $(x, y) = (h^-, h^-)$ is

$$\begin{aligned} E(t) - \frac{|\mathbf{m}(t)|^2}{2\rho(t)} - \frac{|B(h, h, t)|^2}{2} \\ = 1 + A^2 h^2 + \epsilon_1 + 2(\gamma - 1)A^2 h^2 \lambda - 2(\epsilon_1 - \epsilon_3)\gamma\lambda - A^2 h^2 (1 - 2\lambda)^2 \\ - \frac{1}{2}(1 + (\gamma - 1)(A^2 h^2 - (\epsilon_1 - \epsilon_2 - \epsilon_3 + \epsilon_4))\lambda)^2 - \frac{1}{2}(1 + (\gamma - 1)(A^2 h^2 - (\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4))\lambda)^2 \\ = (1 - 2\lambda)\epsilon_1 + 2\lambda\epsilon_3 + 4\lambda A^2 h^2 - \lambda^2 \{4A^2 h^2 \\ + \frac{(\gamma - 1)^2}{2}[A^2 h^2 - (\epsilon_1 - \epsilon_2 - \epsilon_3 + \epsilon_4)]^2 + [A^2 h^2 - (\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4)]^2\}. \end{aligned} \quad (2.7)$$

Now, if $A = 0$ (reduce to the Euler equations) and $\epsilon_1 = \epsilon_3 = 0 < \max\{\epsilon_2, \epsilon_4\}$ and $\epsilon_2 \neq \epsilon_4$, then as long as $\lambda > 0$, the internal energy at (h^-, h^-) is negative!